Modelling COVID-19 Transmission using Mobility and Past Transmission

Yu Kate Chan

School of Public Health, LKS Faculty of Medicine, The University of Hong Kong

Introduction

- COVID-19 transmission is summarised by the **effective reproduction number R_t**.
- **Mobility** is useful in explaining the variability in COVID-19 transmission[1]:

$$\log(R_t) = \log(R_0) - \beta(1 - m_t)$$

- In reality, transmission today would depend on previous days, so adding additional lagged terms (e.g. R_{t-1}) may improve model fit.

Aim & Hypothesis

- **Extend** the model by incorporating additional term(s) involving **past transmission**:

$$\log(R_t) = \log(R_0) - \beta(1 - m_t) + \gamma \log(R_{t-1})$$

$$\log(R_t) = \log(R_0) - \beta(1 - m_t) + \gamma R_{t-1}$$

- Quantify the **relationship** between mobility and transmission across countries.
- Test whether including lagged R_t terms **improves model fit** using both regression and MCMC approaches.

Hypothesis

- Models that include both mobility and past transmission would **better explain** current transmission than mobility-only models.
- Mobility and transmission influence each other in a feedback loop.

Methodology

Data Sources

- **Mobility data** from Google[2] and Apple[3].
- **Daily COVID-19 deaths** from the European Centre for Disease Prevention and Control (ECDC)[4].

Regression Analysis

- Weekly R_t is estimated from smoothed death data to reduce noise.
- **63 Country models**: R_t ~ mobility + lagged R_t, Mobility ~ R_t + lagged mobility
- **Pooled model**: R_t ~ mobility + lagged R_t + country (indicator variable).
- **Vector autoregression** models the two-way relationship between R_t and mobility.
- Model comparison through R^2 for explanatory power and BIC for penalised fit.

MCMC Inference

- Run thousands of iterations to obtain a **posterior distribution** of Rt trajectories.
- **Parameters**: baseline R_0, mobility effect β, lagged R_t effect γ
- Daily simulated R_t is used to predict daily deaths.
- **Likelihood**: compare predicted deaths vs observed deaths.
  - Good model fit → accept parameters; otherwise → reject.
- Compare original vs extended model using DIC for penalised fit.
- Posterior 95% credible intervals used to assess parameter uncertainty.

Results

Country-level Regressions

Transmission models ($\log(R_t)$ as outcome):

- Adding past transmission (R_{t-1} or $\log(R_{t-1})$) improves fit in 47 countries.
- Past transmission is useful in predicting current transmission.

Country	R^2 (no lag)	R^2 (lagged)	BIC (no lag)	BIC (lagged)	Best BIC Model
Brazil	0.03	0.90	-3.4	-83.3	R_{t-1}
Germany	0.33	0.61	21.5	3.1	R_{t-1}
India	0.23	0.51	-16.5	-29.3	$\log(R_{t-1})$
UK	0.43	0.84	15.3	-29.3	R_{t-1}
USA	0.03	0.63	9.2	-24.5	R_{t-1}

Table 1. Country-level regression results for transmission. Comparison of model performance with and without lagged R_t terms. The best model identified for each country.

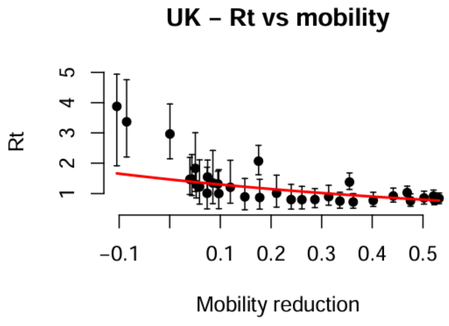


Figure 1. Relationship between R_t and mobility in the UK.

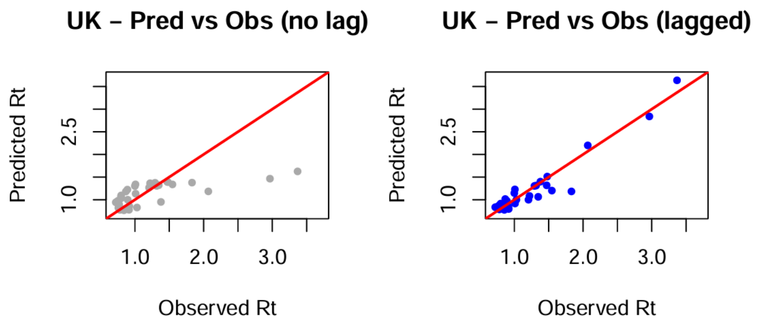


Figure 2. Predicted versus observed R_t in the UK for the baseline and lagged models.

Mobility models (mobility as outcome):

- Adding past mobility improves fit in all countries.
- Mobility is highly autocorrelated: current mobility is correlated with past mobility.

Country	R^2 (no lag)	R^2 (lagged)	BIC (no lag)	BIC (lagged)
Brazil	0.03	0.97	-12.4	-126.0
Germany	0.33	0.95	0.3	-79.8
India	0.23	0.97	-10.9	-108.1
UK	0.43	0.92	-24.4	-86.8
USA	0.03	0.94	2.3	-90.8

Table 2. Country-level regression results for mobility. Comparison of model performance with and without lagged mobility terms.

Vector Autoregression (VAR)

- VAR models describe the two-way relationship between transmission and mobility.
- Granger causality tests show the direction of influence: past transmission is more influential on mobility, whereas the effect of past mobility on transmission is weaker and only observed in certain countries.

Country	BIC	Mobility → R_t (p-value)	R_t → Mobility (p-value)
Brazil	-159.6	0.095	< 0.001
Germany	-61.9	0.045	< 0.001
India	-75.7	0.318	< 0.001
UK	-108.9	0.012	< 0.001
USA	-88.3	0.180	< 0.001

Table 3. VAR results: Testing the feedback loop between transmission and mobility.

Pooled Regressions (All Countries)

- Worldwide results are consistent with individual country findings.

Model	Specification	R^2	BIC
pooled0	log(Rt) ~ mobility + country	0.19	2396.0
pooled1	log(Rt) ~ mobility + Rt-1 + country	0.26	2044.5
pooled_log1	log(Rt) ~ mobility + log(Rt-1) + country	0.25	2088.0
pooled_mobility	mobility ~ log(Rt) + mobility-1 + country	0.96	-4868.1

Table 4. Pooled regression results. Comparison of model specifications across all countries.

MCMC Inference

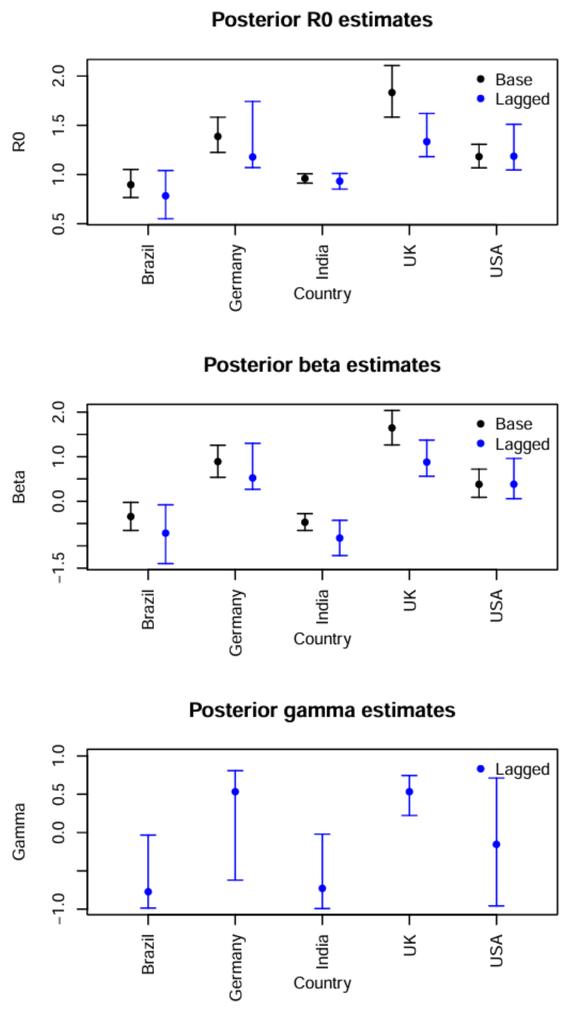


Figure 3. Comparison of posterior parameter estimates for baseline versus lagged models from MCMC inference across five countries.

- Lagged models show small reductions in DIC, indicating that past transmission adds limited explanatory power using MCMC.
- Credible intervals widen in lagged models, reflecting greater uncertainty in parameter estimates.
- Posterior point estimates are similar between baseline and lagged models, suggesting lagged terms do not substantially alter the relationship between R_t and mobility.
- Effects differ by country: in some countries past transmission appears negatively associated with current transmission, which is unexpected.
- Posterior distributions are jointly estimated, which may explain why lagged terms add little improvement to overall model fit.

Country	DIC (no lag)	DIC (lagged)
Brazil	3006.7	3004.5
Germany	1759.7	1753.0
India	2529.7	2527.7
UK	2340.2	2329.8
USA	3429.4	3429.6

Table 5. DIC values for baseline and lagged models fitted using MCMC across five countries.

Summary

- Adding past transmission helps explain current transmission better than mobility-only models, but the model improvement is modest when applied to MCMC simulations.
- Transmission and mobility influence each other in both directions, but past transmission has a stronger effect on mobility than past mobility has on transmission.
- **Research significance**: This approach enables more accurate predictions of R_t and supports more informative public health decisions. The same framework can be applied to other epidemics, with relevant mobility and death data.
- **Future directions**: Investigate how the relationship between mobility and transmission changes before and after control measures were enforced.

References. [1] Nouvellet, P., Bhatia, S., Cori, A. *et al.* (2021). Reduction in mobility and COVID-19 transmission. *Nature Communications* 12, 1090; [2] Google LLC. Google COVID-19 Community Mobility Reports. https://www.google.com/covid19/mobility/; [3] Apple. COVID-19 Mobility Trends Reports. https://www.apple.com/covid19/mobility; [4] European Centre for Disease Prevention and Control (ECDC). Data on the geographic distribution of COVID-19 cases worldwide. https://opendata.ecdc.europa.eu/covid19/casedistribution/csv.